

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a3d467809d3a06397.jsonl`  
- SHA-256: `e2314bfdc5eef7316d567c1ec77516fc19b480862bffd04bfc66c60541107f36`  
- Source modified: `2026-06-06T17:39:00+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:37:55.544Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Scoreboard/score-overlay OCR can be used as the primary temporal-segmentation signal to detect event boundaries (wicket-taking deliveries) in racket-sport-adjacent sports video: frames are sampled, the scorecard ROI is OCR'd, and an event is declared when the recognized score count increments relative to the previous valid reading."

****Source:**** <https://arxiv.org/html/2510.18405> (primary)
****Supporting quote:**** "A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to **refuted=true** if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:37:58.248Z)

I'll verify this claim about scoreboard/score-overlay OCR for event boundary detection in sports video. Let me examine the source.

3. assistant (2026-06-06T17:38:01.393Z)

I'll fetch the primary source to verify the claim's specifics.

4. user (2026-06-06T17:38:09.446Z)

Paper Summary: OCR-Guided Cricket Video Analysis

Sport and Core Focus

This paper addresses ****cricket video analysis****, specifically developing an automated system to identify wicket-taking deliveries and analyze their spatial characteristics.

OCR-Based Temporal Segmentation Method

**** (1) Scoreboard OCR as Segmentation Signal:****

Yes, the approach uses scorecard information for event detection. As stated: "A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance."

**** (2) Frame Sampling and OCR Process:****

The method operates as follows: "The wicket segmentation stage begins by sampling frames from the input broadcast video at fixed temporal intervals, with a default interval of 0.1 s. For each sampled frame, a predefined region of interest (ROI) is used to isolate the scorecard area."

**** (3) Event Declaration Logic:****

Confirmed: "Once such an increment is detected, the corresponding delivery segment is extracted and saved for downstream spatial analysis."

Results Reported

The paper reports strong performance for object detection components:

- ****Pitch detection:**** 99.5% mAP50 with 0.999 precision
- ****Ball detection (transfer learning):**** 99.18% mAP50 with 0.968 precision and 0.978 recall

However, the paper acknowledges: "quantitative evaluation of wicket-event segmentation" remains among future work items, suggesting OCR segmentation accuracy metrics were not formally reported.

5. assistant (2026-06-06T17:38:12.356Z)

The source confirms the mechanics of the claim. Let me note a key qualification surfaced ? the paper does not report quantitative accuracy for the OCR segmentation. Let me verify the sport context and the claim's framing more carefully.

6. user (2026-06-06T17:38:19.271Z)

Cricket Classification & OCR Analysis

****Cricket Sport Classification:****

The document does not explicitly categorize cricket as a "racket sport" or related term. Cricket is simply treated as a sport requiring video analysis.

****OCR Engine Specification:****

The paper does not identify which specific OCR library or engine is employed. It only references "an OCR engine" without naming the implementation.

****Stated Limitations of OCR-Based Detection:****

Yes, the authors acknowledge constraints. The relevant passage states:

> "the dependence on visible scorecards implies that segmentation quality may vary when the broadcast overlay is partially occluded, reformatted, or temporarily absent"

They further note:

> "the system should be interpreted as a practical analytical framework rather than a fully comprehensive cricket intelligence solution"

****Primary Segmentation Signal:****

The paper does position OCR analysis as foundational. The methodology section describes it as the first pipeline stage that "narrows the analysis to event-relevant video portions rather than processing the entire match uniformly." However, the authors do not claim this represents perfect wicket identification?only that it provides practical temporal filtering for downstream processing.

The framework's reliance on consistent broadcast formatting represents its primary operational constraint.

7. assistant (2026-06-06T17:38:22.427Z)

I now have enough to assess the claim. Let me do a quick search to check whether this technique is corroborated by other credible sources (to assess source quality/extraordinariness).

8. user (2026-06-06T17:38:34.283Z)

Web search results for query: "scoreboard OCR score increment event boundary detection sports video segmentation cricket tennis"

Links: [{"title":"Event Detection Based Video Summarization for Sport Video Ph.D. Synopsis", "url":"https://s3-ap-southeast-1.amazonaws.com/gtusitecirculars/uploads/Final%20Synopsis%20(12)_820755.pdf"}, {"title":"Automated Wicket-Taking Delivery Segmentation and Weakness Detection in Cricket Videos Using OCR-Guided YOLOv8 and Trajectory Modeling", "url":"https://arxiv.org/html/2510.18405v1"}, {"title":"An Event Detection based Video Summarization for Cricket ...", "url":"https://www.jetir.org/papers/JETIR2310529.pdf"}, {"title":"(PDF) Automated Wicket-Taking Delivery Segmentation and Weakness Detection in Cricket Videos Using OCR-Guided YOLOv8 and Trajectory Modeling", "url":"https://www.researchgate.net/publication/396747730_Automated_Wicket-Taking_Delivery_Segmentation_and_Weakness_Detection_in_Cricket_Videos_Using_OCR-Guided_YOLOv8_and_Trajectory_Modeling"}, {"title":"Automated Wicket-Taking Delivery Segmentation and Trajectory-Based Dismissal-Zone Analysis in Cricket Videos Using OCR-Guided YOLOv8", "url":"https://arxiv.org/html/2510.18405"}, {"title":"Automated Wicket-Taking Delivery Segmentation and ...", "url":"https://arxiv.org/pdf/2510.18405"}]

Here are the search results for your query about scoreboard OCR, score increment, event boundary detection, and sports video segmentation in cricket and tennis:

Key Findings:

****Cricket Event Detection Using Scoreboards:****

Events in cricket like four runs, six runs and wicket fall can be mined by checking the difference in run and wicket mentioned on the scoreboard. OCR is applied to recognize the score and wicket caption.

****Video Segmentation Process:****

Cricket videos are initially divided into frames used for shot boundary detection, and once shot boundaries are detected, every shot's first frame is preserved as a representative frame and identified as a keyframe, with all further processing applied to these keyframes.

****Modern Approaches:****

Recent systems employ YOLOv8 architecture for pitch and ball detection, combined with optical character recognition (OCR) for scorecard extraction to identify wicket-taking moments. When wicket count increments, the system clips and saves the corresponding video segment containing the complete delivery sequence.

****Scoreboard Localization:****

In broadcast sports video, the scoreboard is attached at a fixed location in the video and generally the scoreboard always exists in all video frames. Methods use Scale Invariant Feature Transform (SIFT) points matching between two frames to localize the scoreboard by computing a robust estimate of the matched point cloud.

****Tennis Ball Tracking:****

Tennis ball tracking using candidate and tracklet-level approaches and random forests for player motion tracking provide insights for sports video analysis.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:38:46.864Z)

The search corroborates the technique broadly: multiple sources (including older work, e.g., JETIR cricket event-detection paper) confirm that scoreboard OCR with run/wicket count differencing is an established method for cricket event mining. The claim's mechanics are well-supported and not extraordinary.

Now let me assess the claim precisely against the verification checklist:

****Claim verbatim:**** "Scoreboard/score-overlay OCR can be used as the primary temporal-segmentation signal to detect event boundaries (wicket-taking deliveries) in racket-sport-adjacent sports video: frames are sampled, the scorecard ROI is OCR'd, and an event is declared when the recognized score count increments relative to the previous valid reading."

Assessment:

1. ****Supported by quote and source?**** Yes, strongly. The paper confirms each mechanical component verbatim: fixed-interval frame sampling (0.1s), a predefined scorecard ROI, OCR of that ROI, and event declaration when wicket count increments relative to the preceding valid score instance. The supporting quote maps exactly. "Primary segmentation signal" is also fair ? the paper positions OCR as the first/foundational pipeline stage.
2. ****Contradicting evidence?**** None. Independent sources corroborate the method as established for cricket.
3. ****Source quality vs. claim strength?**** The claim is modest ("can be used as"), so an arXiv paper plus corroborating literature is sufficient. Not an extraordinary claim.
4. ****Outdated?**** No ? Oct 2025 paper, recent.
5. ****Marketing/cherry-picked?**** No.

One genuine caveat: the paper does NOT quantitatively evaluate the OCR-segmentation accuracy (it lists this as future work), and acknowledges fragility when overlays are occluded/reformatted/absent. But the claim only asserts the method "can be used" and describes the mechanism ? it does not assert any accuracy figure. So these caveats qualify but do not refute the claim as stated. "Racket-sport-adjacent" is the verifier's framing, not the paper's, but it's a defensible loose descriptor and not load-bearing to the technical claim.

The claim is accurate, current, and well-sourced. It is not refuted.

10. user (2026-06-06T17:38:55.099Z)

Structured output provided successfully

11. assistant (2026-06-06T17:39:00.071Z)

Verification complete. The claim is ****not refuted**** (confidence: high).

The primary source (arXiv 2510.18405) supports every mechanical component of the claim verbatim ? fixed-interval frame sampling (0.1s), predefined scorecard ROI, OCR of that ROI, and event declaration on score/wicket count increment versus the preceding valid reading ? and positions OCR as the foundational/primary segmentation stage. An independent cricket paper corroborates the score-differencing technique, confirming it is an established rather than extraordinary method. The claim is modest in strength, current (Oct 2025), and matched by source quality. Genuine caveats exist (the paper defers quantitative segmentation accuracy to future work and flags overlay-occlusion fragility), but since the claim only describes the mechanism without asserting an accuracy figure, these bound reliability rather than refute it.